

ASH 50th Anniversary Review: Human natural killer cells

Natural killer (NK) cells were discovered more than 30 years ago. NK cells are large granular lymphocytes that belong to the innate immune system because unlike T or B lymphocytes of the adaptive or antigen-specific immune system, NK cells do not rearrange T-cell receptor or immunoglobulin genes from their germline configuration. During the past 2 decades there has been a substantial gain in our understanding of what and how NK-cells “see,” lending important insights into their functions and purpose in normal immune surveillance. The most recent discoveries in NK-cell receptor biology have fueled translational research that has led to remarkable results in treating human malignancy.